A3-PS007-TC532

# IPv6-Based IoT Security Monitoring Across Address Rotation

AIORI Hackathon 3 • Technical Solution Proposal


**Team Members**

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**Focus**

Privacy-aware monitoring, address-rotation correlation, and IoT threat detection

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# Problem Description

Clear outline of the problem being addressed

IoT devices such as smart sensors, cameras, and home-automation equipment communicate over IPv6 networks. Privacy mechanisms may cause a device to use changing temporary IPv6 addresses. Monitoring systems that depend heavily on a stable IP address can struggle to associate related activity over time, creating gaps in security visibility.

- **Changing addresses:** temporary IPv6 addresses can change, so IP-only monitoring may lose continuity.
- **Threat visibility:** suspicious activity may be harder to correlate when source addresses change.
- **False positives:** legitimate address rotation may be mistaken for a new or suspicious device.
- **Privacy balance:** security monitoring should avoid unnecessary long-term identification of users or devices.
- **Resource limits:** many IoT devices have limited processing power, so monitoring should be lightweight.

**Problem to solve**

How can a network monitor detect suspicious IoT behavior across IPv6 address rotation without depending only on stable addresses or collecting unnecessary personal data?

# Background

Context needed to understand the problem and proposed optimization

IPv6 supports stateless address autoconfiguration, and temporary addresses can be used to reduce the exposure of a stable interface identifier. This improves privacy, but it also means a security tool cannot assume that one IP address permanently represents one device. The proposed system therefore emphasizes traffic metadata and behavior patterns, while limiting collection of personal information.

**Proposed optimization approach**

- Use authorized network packet capture and extract only relevant metadata, such as timestamps,protocol, packet sizes, and communication frequency.
- Analyze changes in address and communication patterns; do not treat a changed address alone asevidence of an attack.
- Apply transparent detection rules or anomaly scoring to flag unusual behavior for review.
- Protect logs with access controls and retention limits; avoid storing payloads or identifiers unlessrequired for the test.
- Measure detection quality, false-positive rate, processing time, and resource usage against a simplebaseline.

**Scope note:** Correlating activity across changing addresses is probabilistic. The system should report confidence and uncertainty rather than claim that two addresses definitely belong to the same device.

# Literature & RFC Review

Most relevant standards and technical references


**[1] RFC 8981 — Temporary Address Extensions for Stateless Address Autoconfiguration in IPv6 (IETF, 2021).**

Describes temporary IPv6 addresses used for privacy. It explains why address changes must be considered in monitoring design.

**[2] RFC 8200 — Internet Protocol, Version 6 (IPv6) Specification (IETF, 2017).**

Defines IPv6 packet format and protocol behavior, forming the basis for packet parsing and traffic collection.

**[3] RFC 9099 — Operational Security Considerations for IPv6 Networks (IETF, 2021).** Summarizes operational security concerns and practices relevant to deploying and monitoring IPv6 networks.

**[4] NIST SP 800-213 — IoT Device Cybersecurity Guidance for the Federal Government (NIST,2021).**

Provides a risk-based approach to specifying cybersecurity capabilities for IoT devices.

**[5] NIST SP 800-94 — Guide to Intrusion Detection and Prevention Systems (IDPS) (NIST, 2007).** Provides background on network-based monitoring, detection methods, alerting, and evaluation considerations.

**[6] Wireshark User's Guide — Wireshark Foundation.**

Documents a widely used tool for examining packet captures during controlled testing.

# Proposed Solution

Detailed description of the overall solution

Build a lightweight monitoring prototype for a controlled IPv6 IoT test network. The prototype collects authorized traffic, extracts privacy-conscious metadata, analyzes communication behavior, detects anomalies, and displays alerts for a human reviewer.

1. **Collect:** Capture IPv6 traffic from a test network or approved packet dataset. Filter and retain only fields needed for analysis.

2. **Extract features:** Summarize protocol, timestamps, packet/byte counts, flow duration, destination patterns, and communication frequency

3. **Analyze rotation:** Record address-change events and compare time-windowed behavior. Do not rely on the address alone to infer device identity.

4. **Detect anomalies:** Use baseline rules or an anomaly-detection model to flag unusual traffic volume, unexpected destinations, repeated connection attempts, or protocol deviations.

5. **Alert and report:** Show severity, time, evidence, and confidence on a dashboard. Keep a human reviewer in the loop.

6. **Protect privacy:** Minimize stored data, restrict log access, set retention periods, and use pseudonymous identifiers only where necessary.

**Expected output:** A dashboard showing traffic status, address-change events, detected anomalies, and a short explanation for each alert. Automated blocking is out of scope for the initial prototype; it may be considered only after safe validation.

# Solution Optimization

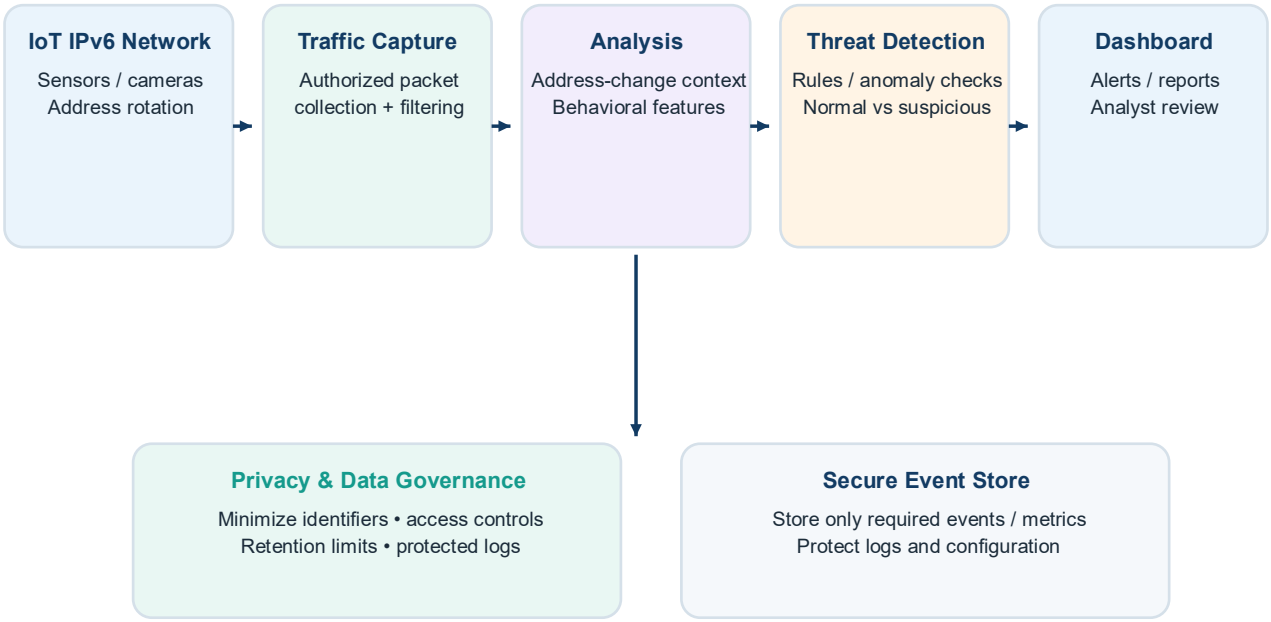
Before-and-after comparison of proposed enhancements

| Area | Before / Baseline | After / Proposed Enhancement |
|---|---|---|
| **Address handling** | Monitoring relies heavily on a fixed IP address. | Use time-windowed traffic features and address-change context; report correlation confidence. |
| **Detection** | Static checks may miss behavior changes. | Combine transparent rules with behavior-based anomaly indicators. |
| **False alerts** | Address rotation itself may trigger alerts. | Treat rotation as context, not as an attack by itself; validate against normal traffic. |
| **Data collection** | Potentially collects more packet detail than required. | Filter traffic and minimize payload or personal-data collection. |
| | | |
| **Performance** | All traffic may be processed in the same way. | Use flow summaries, filtering, and bounded time windows to reduce overhead. |
| **Reporting** | Logs may be difficult to interpret. | Provide severity, timestamp, observed evidence, and recommended review action. |
| **Evaluation** | No consistent baseline measurement. | Compare detection rate, false-positive rate, latency, and resource usage. |

**Validation principle:** Improvements are targets, not guaranteed results. Compare the optimized prototype with the baseline using the same test traffic and documented metrics.

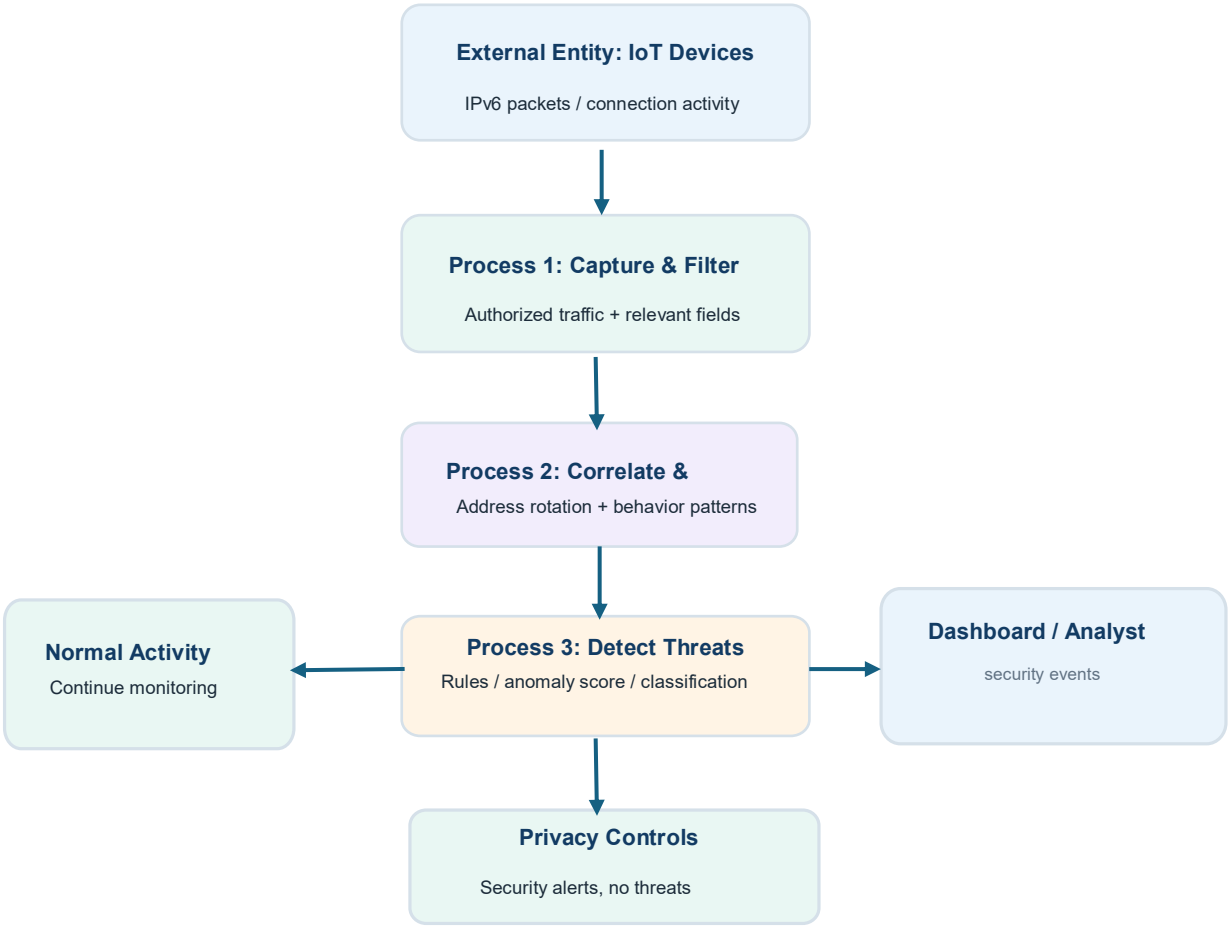
# Solution Architecture & Design

Visual overview of system components and optimization process



# Data Flow Diagram

Level 1 data flow for IPv6-based IoT security monitoring




**Flow summary:** IPv6 packets enter the capture process; filtered metadata is analyzed for address rotation and behavior patterns; the detection engine separates normal activity from suspicious events; alerts and summaries are sent to the dashboard. Minimal event records are stored securely under privacy controls.

# Deployment Timeline

Milestones and methodology

| Phase | Activities | Deliverable |
|---|---|---|
| **1. Requirements** | Review RFCs, define threat scenarios, privacy boundaries, and test scope. | Requirements and evaluation plan |
| **2. Design** | Define capture, feature extraction, analysis, storage, and dashboard modules. | Architecture and data-flow diagram |
| **3. Build** | Set up controlled IPv6 traffic source; implement capture and metadata extraction. | Traffic collection prototype |
| **4. Detect** | Implement baseline rules and behavior-based anomaly indicators. | Detection module |
| **5. Optimize** | Tune filtering and time windows; review privacy and logging controls. | Optimized prototype |
| **6. Validate** | Test normal traffic, address rotation, and safe simulated anomalies; record metrics. | Test report and performance results |
| **7. Demo** | Run end-to-end demonstration and explain limitations. | Dashboard demo and final report |

**Suggested evaluation metrics:** detection rate/recall, precision, false-positive rate, alert latency, CPU/memory usage, and amount of data retained. Use repeatable test cases and clearly document assumptions.

# Bibliography

Standards and documentation used to frame the solution


[1] IETF, “Privacy Extensions for Stateless Address Autoconfiguration in IPv6,” RFC 8981, 2021. Available: https://www.rfc-editor.org/rfc/rfc8981.html

[2] S. Deering and R. Hinden, “Internet Protocol, Version 6 (IPv6) Specification,” RFC 8200, IETF, 2017. Available: https://www.rfc-editor.org/rfc/rfc8200.html

[3] E. Vyncke et al., “Operational Security Considerations for IPv6 Networks,” RFC 9099, IETF, 2021. Available: https://www.rfc-editor.org/rfc/rfc9099.html

[4] NIST, “IoT Device Cybersecurity Guidance for the Federal Government: Establishing IoT Device Cybersecurity Requirements,” NIST SP 800-213, 2021. Available: https://doi.org/10.6028/NIST.SP.800-213

[5] K. Scarfone and P. Mell, “Guide to Intrusion Detection and Prevention Systems (IDPS),” NIST SP 800-94, 2007. Available: https://doi.org/10.6028/NIST.SP.800-94

[6] NIST, “Foundational Cybersecurity Activities for IoT Device Manufacturers,” NISTIR 8259, 2020. Available: https://doi.org/10.6028/NIST.IR.8259

[7] Wireshark Foundation, “Wireshark User’s Guide.” Available: https://www.wireshark.org/docs/


Accessed for project planning on October 13, 2026. Standards and documentation support the design rationale; they do not themselves validate the prototype’s performance.

# Thank You

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IPv6-Based IoT Security Monitoring Across Address Rotation

**Key idea**
Monitor behavior, not just addresses.
Detect suspicious activity while respecting privacy.

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